TSEGA ABEBE

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EDUCATION

Yale-NUS College, National University of Singapore — Singapore
B.Sc. (Hons) Mathematical, Computational and Statistical Sciences | Major **GPA**: 4.5/5 — **Aug 2021 – May 2025**
Relevant Coursework: Data Structures and Algorithms, Software Engineering, Machine Learning, Programming for Data Science, Database Management Systems, Programming Operating Systems, Software Verification and Validation, Computer Networks

Yale University — New Haven, CT
Exchange program, B.Sc. Computer Science — **Jan 2024 – May 2024**
Relevant Coursework: AI for Future Presidents, Full Stack Web Programming, Object-Oriented Programming, Human-Computer Interaction (HCI)

PROFESSIONAL SKILLS

- Languages: Python, C/C++, JavaScript (Node.js, React), Java, HTML, CSS, SQL, OCaml, R, LaTeX
- Cloud & DevOps: Microsoft Azure, Kubernetes (OpenShift), Docker, Git/GitHub, Linux, Rest API, CI/CD, Copilot Studio
- AI & ML: TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, Retrieval-Augmented Generation, LangChain,Vector Search
- Databases & Tools: PostgreSQL, Qdrant, MongoDB, Redis, ETL Pipelines, Postman, Figma, Power Platform

EXPERIENCE

Robert Bosch (SEA), Singapore — **Aug 2025 – Present**
AI/Cloud Engineer — *(Previously Intern, May 2024 – Apr 2025)*

- Architected and deployed an **enterprise-grade RAG** system integrating on-premises data sources and gateways, Azure AI Search, LangChain, and Teams to power AI agents automating manual workflows for over **200+** employees**.**
- Designed and built the full ETL → chunking → embedding → indexing pipeline for agentic document retrieval with **permission filtering** for over **10K** enterprise files, achieving sub-second latency and reducing search time by **~70%** based on user testing**.**
- Implemented **multi-agent** orchestration using MCP servers, self-hosted N8N, Azure services and Copilot Studio, automating request routing and enforcing audit logging for enterprise compliance; resulting in **~40%** faster processing.
- Partnered with cross-functional teams to deliver **scalable prototypes** and technical documentation aligned with enterprise AI strategy.

Summer Research Program, Yale-NUS College — **May 2023 – Jul 2023**
Software Developer Intern

- Collaborated in constructing a publicly accessible database of over **150** spectral line detection for galaxies (Website).
- Engineered a responsive and elegant pagination component for the database web interface. Enhanced user experience by dynamically loading galaxy datasets on demand, optimizing website speed by reducing initial page load time from **~8** seconds to **~2** seconds.
- Integrated features using JavaScript, enabling users to export search results in CSV format.

eGROCERY Web Application, Yale-NUS College — **Jan 2023 – Apr 2023**
Software Architect

- Authored an extensive SRS document for our final project following IEEE standards, detailing the software's functionalities and serving as the technical blueprint that streamlined development and reduced scope ambiguity.
- Architected a robust backend infrastructure with Python (Flask framework), focusing on performance, scalability, and security.
- Designed a MongoDB database and implemented a comprehensive set of **RESTful API** endpoints for dynamic inventory and user management designed to scale to **100,000** products and users; Rigorously tested with Postman for reliability and performance.
- Led the project with daily Scrum meetings, followed Agile methodology and used GitHub for CI/CD processes (GitHub).

PROJECTS

Customer Churn Prediction Using Supervised Learning in R (Github)

- Engineered and deployed ensemble ML pipeline combining Logistic Regression, Random Forest, and XGBoost with SMOTE oversampling, achieving 68% AUC on heavily imbalanced dataset through hyperparameter tuning and stacked meta-learning.
- Developed domain-specific features via feature engineering, improving model precision by **5%**.

Santorini Game Implementation with Design Patterns (Github)

- Built a modular CLI version of the Santorini board game using Strategy, Observer, and Command design patterns, featuring AI opponents with heuristic-based decision-making, a scoring system, and undo/redo functionality.

Gesture-Controlled Feedback System (Demo)

- Led a team of four in developing an interactive system using Kinect Azure motion sensor to capture hand gestures, allowing students to rate each lecture on a scale of 1-10, providing real-time feedback to professors. Won Class Choice Award for my HCI class at Yale.

LEADERSHIP & OTHER EXPERIENCE

The Afro Society, Yale-NUS College | President — **Aug 2022 – Present**

- Spearheaded cultural, diversity, and professional engagement initiatives, including **8+** panel discussions and workshops with **50+** attendees, driving an **increase in community participation** among International students and professionals.

College Digital Archives, Yale-NUS College | Student Associate Senior Lead — **Jan 2022 – May 2025**

- Manage metadata for **56,000+** digital archive items, overseeing creation, validation, and quality control, while supervising and training a team of **6** student associates in Agile workflows.